

When Does Aggregating Explanations Work?

OrganAMNIST Results

Jinhua Xu, Davide Anguita, Fabio Roli, Jing Yuan, and Luca Oneto

This report is generated from the structured result files listed in `RESULT_MANIFEST.json`. It is organised by model and experiment and retains incomplete and historical snapshots as archival tables.

Historical and superseded result tables

A DenseNet-121 (archived)

Table 1: Real-checkpoint explainer compatibility pilot for OrganAMNIST–DenseNet-121. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	DenseNet121	passed	blocked	Saliency	false	false	1		1				RuntimeError: Deter- min- istic repeat check failed
OrganAMNIST	DenseNet121	passed	blocked	InputXGradient	false	false	1		1				RuntimeError: Deter- min- istic repeat check failed
OrganAMNIST	DenseNet121	passed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0875516.6774e+09		
OrganAMNIST	DenseNet121	passed	blocked	GuidedBackprop	false	false	1		1				RuntimeError: Deter- min- istic repeat check failed
OrganAMNIST	DenseNet121	passed	blocked	Deconvolution	false	false	1		1				RuntimeError: Deter- min- istic repeat check failed
OrganAMNIST	DenseNet121	passed	blocked	FeatureAblation	true	true	1	true	1				
OrganAMNIST	DenseNet121	passed	blocked	Occlusion	true	true	1	true	1				
OrganAMNIST	DenseNet121	passed	blocked	DeepLift	true	true	1	true	1	1	0.0642516.6008e+08		
OrganAMNIST	DenseNet121	passed	blocked	GradientShap	true	true	1	true	1				
OrganAMNIST	DenseNet121	passed	blocked	DeepLiftShap	true	true	1	true	1	1	0.05542 1.7365e+09		
OrganAMNIST	DenseNet121	passed	blocked	LRP	true	true	1	true	1				

B EfficientNet-B0 (archived)

Table 2: Real-checkpoint explainer compatibility pilot for OrganAMNIST–EfficientNet-B0. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	infinite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	EfficientNet B0	Failed	blocked	Saliency	true	true	1	true	1	1	0.3751473	8026e+08	
OrganAMNIST	EfficientNet B0	Failed	blocked	InputXGradient	true	true	1	true	1	1	0.2457953	8267e+08	
OrganAMNIST	EfficientNet B0	Failed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0368384	5561e+09	
OrganAMNIST	EfficientNet B0	Failed	blocked	GuidedBackprop	true	true	1	true	1				
OrganAMNIST	EfficientNet B0	Failed	blocked	Deconvolution	true	true	1	true	1				
OrganAMNIST	EfficientNet B0	Failed	blocked	FeatureAblation	true	true	1	true	1				
OrganAMNIST	EfficientNet B0	Failed	blocked	Occlusion	true	true	1	true	1				
OrganAMNIST	EfficientNet B0	Failed	blocked	DeepLift	true	true	1	true	1	1	0.0226722	7869e+08	
OrganAMNIST	EfficientNet B0	Failed	blocked	GradientShap	true	true	1	true	1				
OrganAMNIST	EfficientNet B0	Failed	blocked	DeepLiftShap	true	true	1	true	1	1	0.0238366	5415e+08	
OrganAMNIST	EfficientNet B0	Failed	blocked	LRP	false								RuntimeError: No expla- nation batch candi- date passed

C MobileNetV3-Large (archived)

Table 3: Real-checkpoint explainer compatibility pilot for OrganAMNIST–MobileNetV3-Large. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	infinite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	MobileNetV3-Large	failed	blocked	Saliency	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	InputXGradient	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0211382	4.103e+09	
OrganAMNIST	MobileNetV3-Large	failed	blocked	GuidedBackprop	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	Deconvolution	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	FeatureAblation	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	Occlusion	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	DeepLift	true	true	1	true	1	1	0.0173722	2.2837e+08	
OrganAMNIST	MobileNetV3-Large	failed	blocked	GradientShap	true	true	1	true	1				
OrganAMNIST	MobileNetV3-Large	failed	blocked	DeepLiftShap	true	true	1	true	1	1	0.01657	4.9993e+08	
OrganAMNIST	MobileNetV3-Large	failed	blocked	LRP	false								RuntimeError: No expla- nation batch candi- date passed

D ResNet-18 (archived)

Table 4: Real-checkpoint explainer compatibility pilot for OrganAMNIST-ResNet-18. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	ResNet-18	passed	blocked	Saliency	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-18	passed	blocked	InputXGradient	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-18	passed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0329511	1.2722e+09	
OrganAMNIST	ResNet-18	passed	blocked	GuidedBackprop	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-18	passed	blocked	Deconvolution	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-18	passed	blocked	FeatureAblation	true	true	1	true	1				
OrganAMNIST	ResNet-18	passed	blocked	Occlusion	true	true	1	true	1				
OrganAMNIST	ResNet-18	passed	blocked	DeepLift	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-18	passed	blocked	GradientShap	true	true	1	true	1				
OrganAMNIST	ResNet-18	passed	blocked	DeepLiftShap	true	true	1	true	1	1	0.01683	4.4585e+08	
OrganAMNIST	ResNet-18	passed	blocked	LRP	true	true	1	true	1				

E ResNet-50 (archived)

Table 5: Real-checkpoint explainer compatibility pilot for OrganAMNIST-ResNet-50. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	ResNet-50	failed	blocked	Saliency	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-50	failed	blocked	InputXGradient	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-50	failed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0710094	5233e+09	
OrganAMNIST	ResNet-50	failed	blocked	GuidedBackprop	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-50	failed	blocked	Deconvolution	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-50	failed	blocked	FeatureAblation	true	true	1	true	1				
OrganAMNIST	ResNet-50	failed	blocked	Occlusion	true	true	1	true	1				
OrganAMNIST	ResNet-50	failed	blocked	DeepLift	false	false	1		1				RuntimeError: Deterministic repeat check failed
OrganAMNIST	ResNet-50	failed	blocked	GradientShap	true	true	1	true	1				
OrganAMNIST	ResNet-50	failed	blocked	DeepLiftShap	true	true	1	true	1	1	0.02656	1.2475e+09	

dataset	model	result status	gate status	method	passed	determine repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	ResNet-50	failed	blocked	LRP	false								RuntimeError: No expla- nation batch candi- date passed

F ViT-B/16 (archived)

Table 6: NAIVE results for OrganAMNIST–ViT-B/16. This table is a historical snapshot included for archival comp the source file.

dataset	model	architecture	split	sample count	method	F	Fbar	C	Cbar	R F g	R Fbar g	R C g	R Cbar g	R F p	R Fbar p	R C p	R Cbar p	R F s	R Fbar s	R C s	R Cbar s	R F a	R Fbar a	R C a	R Cbar a
organamnist	organamnist-vit-b16	organamnist-vit-b16	test	17778	best_ind	0.00071	0.246484	0.075824	0.276859	0.018	-	0.007031	-	0.010462	-	0.000169	-	0.009337	-	-	-	0.100067	-	0.098324	-
organamnist	organamnist-vit-b16	organamnist-vit-b16	test	17778	simpleavg	0.022893	0.267072	0.062662	0.298684	0.019406	-	0.018787	-	0.010294	-	0.010744	-	0.0117	0.017887	0.009112	-	0.024806	-	0.019294	-
organamnist	organamnist-vit-b16	organamnist-vit-b16	test	17778	borda	0.020531	0.28434	0.056981	0.316402	0.014006	-	0.014625	-	0.010575	-	0.013612	-	0.005569	0.014794	0.005175	-	0.036956	-	0.036056	-
organamnist	organamnist-vit-b16	organamnist-vit-b16	test	17778	kemeny	0.019406	0.29019	0.056981	0.321971	0.015412	-	0.01665	-	0.008325	-	0.013556	-	0.0045	0.018956	0.00225	-	0.03105	-	0.026268	-
organamnist	organamnist-vit-b16	organamnist-vit-b16	test	17778	rrf	0.0198	0.284959	0.057374	0.317021	0.015637	-	0.017325	-	0.010462	-	0.014625	-	0.005906	0.018562	0.00585	-	0.031612	-	0.029193	-
organamnist	organamnist-vit-b16	organamnist-vit-b16	test	17778	schulze	0.020418	0.287828	0.057993	0.319609	0.014906	-	0.0162	-	0.007987	-	0.011419	-	0.004669	0.02025	0.003825	-	0.031331	-	0.028125	-
											0.116774		0.165429		0.094161		0.134942			0.007144		0.144223			

Table 7: NAIVE results for OrganAMNIST-ViT-B/16. This table is a historical snapshot included for archival comparison; it reports only the dataset-model rows present in the source file.

scope	dataset	model	group	comparison cells	method	wins	vs best better	vs best equal	vs best worse	vs sim- pleavg better	vs sim- pleavg equal	vs sim- pleavg worse
dataset_model	organamnist	vit-b16	mult	20	best_individual							
dataset_model	organamnist	vit-b16	mult	20	simpleavg	4	11	0	9			
dataset_model	organamnist	vit-b16	mult	20	borda	1	10	0	10	10	0	10
dataset_model	organamnist	vit-b16	mult	20	kemeny	0	9	0	11	10	0	10
dataset_model	organamnist	vit-b16	mult	20	rrf	5	9	1	10	12	0	8
dataset_model	organamnist	vit-b16	mult	20	schulze	2	9	0	11	11	0	9
dataset_model	organamnist	vit-b16	quality	4	best_individual							
dataset_model	organamnist	vit-b16	quality	4	simpleavg	0	0	0	4			
dataset_model	organamnist	vit-b16	quality	4	borda	0	0	0	4	0	0	4
dataset_model	organamnist	vit-b16	quality	4	kemeny	0	0	0	4	0	0	4
dataset_model	organamnist	vit-b16	quality	4	rrf	0	0	0	4	0	0	4
dataset_model	organamnist	vit-b16	quality	4	schulze	0	0	0	4	0	0	4
dataset_model	organamnist	vit-b16	robustness	16_all	best_individual							
dataset_model	organamnist	vit-b16	robustness	16_all	simpleavg	4	11	0	5			
dataset_model	organamnist	vit-b16	robustness	16_all	borda	1	10	0	6	10	0	6
dataset_model	organamnist	vit-b16	robustness	16_all	kemeny	0	9	0	7	10	0	6
dataset_model	organamnist	vit-b16	robustness	16_all	rrf	5	9	1	6	12	0	4
dataset_model	organamnist	vit-b16	robustness	16_all	schulze	2	9	0	7	11	0	5
dataset_model	organamnist	vit-b16	gaussian	4	best_individual							
dataset_model	organamnist	vit-b16	gaussian	4	simpleavg	2	4	0	0			
dataset_model	organamnist	vit-b16	gaussian	4	borda	0	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	gaussian	4	kemeny	0	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	gaussian	4	rrf	2	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	gaussian	4	schulze	0	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	salt-pepper	4	best_individual							
dataset_model	organamnist	vit-b16	salt-pepper	4	simpleavg	0	3	0	1			

scope	dataset	model	group	comparison cells	method	wins	vs best better	vs best equal	vs best worse	vs sim- pleavg better	vs sim- pleavg equal	vs sim- pleavg worse
dataset_model	organamnist	vit-b16	dataset	pepper	borda	1	4	0	0	4	0	0
dataset_model	organamnist	vit-b16	dataset	pepper	kemeny	0	3	0	1	3	0	1
dataset_model	organamnist	vit-b16	dataset	pepper	rrf	3	3	1	0	4	0	0
dataset_model	organamnist	vit-b16	dataset	pepper	schulze	0	3	0	1	3	0	1
dataset_model	organamnist	vit-b16	dataset	spoke	best_individual	4						
dataset_model	organamnist	vit-b16	dataset	spoke	simpleavg	2	4	0	0			
dataset_model	organamnist	vit-b16	dataset	spoke	borda	0	3	0	1	0	0	4
dataset_model	organamnist	vit-b16	dataset	spoke	kemeny	0	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	dataset	spoke	rrf	0	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	dataset	spoke	schulze	2	3	0	1	2	0	2
dataset_model	organamnist	vit-b16	dataset	universal	best_individual	4						
dataset_model	organamnist	vit-b16	dataset	universal	simpleavg	0	0	0	4			
dataset_model	organamnist	vit-b16	dataset	universal	borda	0	0	0	4	4	0	0
dataset_model	organamnist	vit-b16	dataset	universal	kemeny	0	0	0	4	3	0	1
dataset_model	organamnist	vit-b16	dataset	universal	rrf	0	0	0	4	4	0	0
dataset_model	organamnist	vit-b16	dataset	universal	schulze	0	0	0	4	4	0	0
overall			all	20	best_individual	8						
overall			all	20	simpleavg	4	11	0	9			
overall			all	20	borda	1	10	0	10	10	0	10
overall			all	20	kemeny	0	9	0	11	10	0	10
overall			all	20	rrf	5	9	1	10	12	0	8
overall			all	20	schulze	2	9	0	11	11	0	9
overall			quality	4	best_individual	4						
overall			quality	4	simpleavg	0	0	0	4			
overall			quality	4	borda	0	0	0	4	0	0	4
overall			quality	4	kemeny	0	0	0	4	0	0	4
overall			quality	4	rrf	0	0	0	4	0	0	4
overall			quality	4	schulze	0	0	0	4	0	0	4
overall			robustness	6	best_individual	4						
overall			robustness	6	simpleavg	4	11	0	5			
overall			robustness	6	borda	1	10	0	6	10	0	6
overall			robustness	6	kemeny	0	9	0	7	10	0	6
overall			robustness	6	rrf	5	9	1	6	12	0	4
overall			robustness	6	schulze	2	9	0	7	11	0	5
overall			gaussian	4	best_individual	4						
overall			gaussian	4	simpleavg	2	4	0	0			
overall			gaussian	4	borda	0	3	0	1	2	0	2
overall			gaussian	4	kemeny	0	3	0	1	2	0	2

scope	dataset	model	group	comparison cells	method	wins	vs best better	vs best equal	vs best worse	vs sim- pleavg better	vs sim- pleavg equal	vs sim- pleavg worse
overall			gaussian	4	rrf	2	3	0	1	2	0	2
overall			gaussian	4	schulze	0	3	0	1	2	0	2
overall			salt_pepper	4	best_individual							
overall			salt_pepper	4	simpleavg	0	3	0	1			
overall			salt_pepper	4	borda	1	4	0	0	4	0	0
overall			salt_pepper	4	kemeny	0	3	0	1	3	0	1
overall			salt_pepper	4	rrf	3	3	1	0	4	0	0
overall			salt_pepper	4	schulze	0	3	0	1	3	0	1
overall			speckle	4	best_individual							
overall			speckle	4	simpleavg	0	4	0	0			
overall			speckle	4	borda	0	3	0	1	0	0	4
overall			speckle	4	kemeny	0	3	0	1	2	0	2
overall			speckle	4	rrf	0	3	0	1	2	0	2
overall			speckle	4	schulze	2	3	0	1	2	0	2
overall			adversarial	4	best_individual							
overall			adversarial	4	simpleavg	0	0	0	4			
overall			adversarial	4	borda	0	0	0	4	4	0	0
overall			adversarial	4	kemeny	0	0	0	4	3	0	1
overall			adversarial	4	rrf	0	0	0	4	4	0	0
overall			adversarial	4	schulze	0	0	0	4	4	0	0

Table 8: Real-checkpoint explainer compatibility pilot for OrganAMNIST-ViT-B/16. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	infinite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
OrganAMNIST	ViT-B/16	passed	passed	Saliency	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	InputXGradient	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	IntegratedGradients	true	true	1	true	1	1	0.1419076	6.6673e+09	
OrganAMNIST	ViT-B/16	passed	passed	FeatureAblation	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	Occlusion	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	GradientShap	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	CheferTransformerAttribution	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	PartialLRP	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	FullLRP	true	true	1	true	1				
OrganAMNIST	ViT-B/16	passed	passed	GradientAttentionRollout	true	true	1	true	1	1	0.0434683	3.3843e+09	
OrganAMNIST	ViT-B/16	passed	passed	AttentionGradCAM	true	true	1	true	1	1	0.0378393	3.3106e+09	